CRIMEVISIONAI

Define Problem Statements

Folder 1 | Problem definition | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Primary problem statement

Investigators and forensic reviewers may need to examine audio recordings, photographs, and video clips separately, manually note observations, and reconstruct a chronology across multiple files. This fragmented workflow can increase review effort and make it harder to retrieve relevant observations quickly. CrimeVisionAI explores a unified workspace that assists with evidence intake, modality-specific analysis, timestamped event organization and case-level question answering.

2. Stakeholders and needs

Stakeholder	Need	Design response
Investigator / reviewer	Review multiple evidence types in one case context	Case workspace, evidence list, analysis results and timeline.
Forensic analyst	Inspect source-linked machine outputs without confusing them with facts	Display transcripts, detections, captions and generated interpretations separately.
Project evaluator	Understand what works and how it was tested	Documented architecture, feature status, reproducible setup and test evidence.
Data / system administrator	Protect evidence and keep the application maintainable	Local storage baseline, controlled access in future hardening, error handling and documented limits.

3. Problem dimensions

Dimension	Current challenge	Prototype objective
Format fragmentation	Audio, image and video require different tools	Provide one case-scoped intake and analysis workflow.
Manual review burden	Transcribing speech and scanning visual content take time	Automate first-pass transcription, detection, captioning and candidate event extraction.
Chronology reconstruction	Observations may be distributed across evidence files	Store timestamped events and provide a common timeline view.
Retrieval	Finding relevant observations may require repeated manual inspection	Provide investigator Q&A over available case analysis.
Reliability	AI outputs can be incomplete, biased or hallucinated	Expose uncertainty and validate answers against recorded evidence outputs.

4. Scope boundaries

Included in the known baseline: Flask web application, case/evidence records, upload and media routes, audio transcription, image/video analysis, timeline-oriented outputs and case chat. Excluded or not yet verified: certified forensic acquisition, chain-of-custody compliance, speaker diarization, fully reliable cross-modal entity identity, automated legal conclusions, production-grade multi-user authorization, and guaranteed real-time processing.

5. Success criteria

Success means a reviewer can run the application locally, create or select a case, upload supported evidence, inspect analysis outputs, locate timestamped events, and use Q&A; with visible limitations. Quantitative accuracy, latency and throughput targets must be set and measured separately; no unmeasured metric is claimed here.